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Floragenist® Rose

The Skin Perfection Elixir

Global Except US

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SUMMARY FILE

Floragenist Rose	BC10181
Origin - Description	Natural Rose bud extract produced by AquaGenesis patented extraction technology, enriched in flavonol glycosides and tannins
Regulatory data	
INCI	Maltodextrin (and) Cocos Nucifera (Coconut) Fruit Extract (and) Rosa Damascena Flower Extract
China	All components are listed in Inventory of Existing Cosmetic Ingredient in China (IECIC)
CAS# EINECS	9050-36-6; 8001-31-8; 90106-38-0 232-940-4; 232-282-8; 290-260-3
Appearance	Light beige to light brown powder, potentially pinky
Natural labels	Raw material conforms to Cosmos standards of natural and organic cosmetics
Naturalness content (ISO 16128)	100% from natural origin
Cosmetic use	
Properties	• Helps keratinocyte differentiation for improved epidermis architecture • Helps to strengthen cell-cell adhesion complexes for epidermal and *stratum corneum* cohesion • Helps to protect skin cells against oxidative stress and UVB-mediated pro-inflammatory response • Contributes to skin perfection, luminosity and redness improvement, skin homogeneity and appearance of overall skin health
Applications	• Face care, Skin care, Skin perfection, Radiance-enhancing care
Formulation data	
Concentration of use	0.4%
Solubility	Soluble in water
Incorporation method	Solubilize in water up to 85°C or at room temperature for cold processing
Optimal pH	4-8



Floragenist range
Next-generation botanical extracts reinvented

Floragenist Rose captures the timeless beauty of Damask Rose while elevating it with the precision of modern science.

Today's consumers want both: the reassurance of authentic botanicals and the confidence of clinically supported results. With AquaGenesis technology, our next generation coconut water powered extraction technology, each rosebud is transformed into a supercharged floral elixir rich in naturally occurring flavonol glycosides and tannins.

Water derived ingredients play a central role in cosmetic science and skincare applications, contributing to skin hydration, comfort, and sensory attributes in beauty formulations. In the cosmetic and personal care industry, increasing attention has been given to naturally sourced ingredients that support skin care routines while meeting consumer expectations for sustainability and efficacy [Mendake *et al*, 2025].

Among coconut derived ingredients, coconut water has gained particular interest in recent years. Coconut water, the liquid endosperm of Cocos nucifera, is characterized by its hydrating properties and its natural content in sugars, vitamins, and minerals, which underpin its use in cosmetic and skin care applications [Mu *et al*, 2024]. Consequently, coconut water is increasingly incorporated into cosmetic formulations and beauty products, where it is valued as a refreshing, naturally derived ingredient contributing to skin and hair moisturization [Mendake *et al*, 2025].

What if we could unlock its hidden potential and harness the power of nature to create something truly extraordinary?

In 2024, 22% of US Beauty and Personal Care users reported finding it hard to trust new ingredients in products, making it essential to reintroduce familiar elements with a scientific twist [Mintel, 2024a; Mintel, 2024b]. Additionally, 36% of US facial skincare users seek natural or organic ingredients and 51% UK product launches carry natural/organic claims [Mintel, 2025a; Mintel GNPD, 2025].

This dual expectation has paved the way for next generation floral extracts powered by AquaGenesis technology designed to unlock a higher concentration of active molecules in natural extracts, combining tradition and innovation.

Let's introduce Floragenist range: aquatic elixirs obtained by AquaGenesis-patented technology using coconut water as natural solvent to enhance phytochemical content. Floragenist Rose is part of this innovative range, alongside Floragenist Blossom.



Floragenist Rose Damask Rose, perfected by science

Rose extracts in modern cosmetics

Rose flower extracts increased by 172% in new product launches in skincare over the past decade, indicating a significant market interest in this botanical ingredient [Mintel GNPD, 2024].

Timeless allure = The rose stands as an enduring symbol in cosmetics, embodying feminity and delicate power of nature

The timeless allure of the rose in cosmetics goes far beyond its aesthetic qualities. Indeed, the rose is known for its regenerating and soothing properties on the skin, making it a sought-after ingredient in skincare products [Jwala Jayapal *et al*, 2025]. For centuries, the rose has been revered as a symbol of eternal beauty and romance. Ancient civilizations cherished the rose, including Cleopatra, who used its petals to maintain radiant and glowing skin.

A myriad of exquisite extracts to pamper and beautify = Rosewater, Rosehip oil, Rose absolute, Rose extracts, Rose stem cells

Today, modern research reaffirms the potential of the rose for skin perfection. Its extracts, rich in phytochemicals, help to protect the skin from environmental damage, reduce inflammation, and promote cellular regeneration [Zhao *et al*, 2025]. The diversity of rose-derived products, such as rosewater, rosehip oil, rose absolute, rose extracts, and rose stem cells, allows the formulation of skincare tailored to every skin type.

By merging ecological innovation with the exotic charm of the rose, scientists create cosmetic products that honor tradition while meeting modern demands for sustainability and efficacy [Mintel, 2025b]. The rose, with its enchanting fragrance and beneficial properties, continues to be a cornerstone in the beauty industry, offering natural and luxurious solutions to pamper the skin and reveal its inherent beauty.

The quest for skin perfection: true beauty comes from within

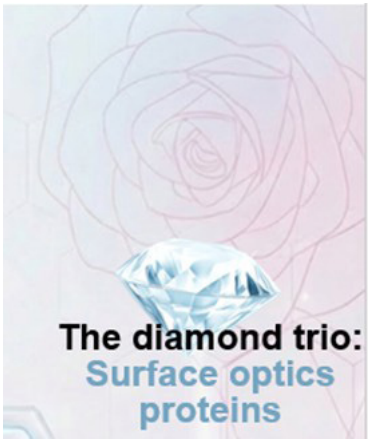
Skin perfection is the most universally desired human feature where healthy skin appearance can be related to person's health and attractiveness perception.Skin perfection is a multifactorial and sensorial skin benefit quest. "Skin perfection" mainly referred to "skin radiance" and "skin glow" to offer smoothness, clarity, and even tone of the skin [Humphrey *et al*, 2021].

"Skin radiance" is a perception that refers to the healthy, luminous appearance of the skin. This perception can be related to the maintaining of the skin integrity and structure which will help to prevent water loss and to protect against external aggression through optimal cell cohesion. When this cohesion is compromised, skin can become dry, irritated (redness) or dull, affecting skin microrelief and skin radiance [Zhang L *et al*, 2024]. In addition, the spatial pattern of redness signals on facial skin has been shown to relate to observers' perceptions of skin health and stress [Omotezako *et al*, 2024].

"Skin glow" also refers to a perception of the skin radiance, hydration, and overall health through reflection of light [Goodman *et al*, 2024].

In summary, skin homogeneity, redness, and luminosity are closely linked to the perception and presence of skin imperfections and collectively contribute to a healthier and more attractive appearance.

Behind skin radiance: surface “Optics” proteins



Involucrin, filaggrin, and corneodesmosin are essential epidermal proteins that together ensure the epidermis structural integrity and robust outermost layer of the skin.

Smoothness – Involucrin (IVL)
Involucrin plays a crucial role in skin biology by serving as one of the key proteins involved in the formation of the cornified cell envelope, which is essential for the integrity and barrier function of the epidermis. As keratinocytes mature and migrate towards the surface, involucrin is incorporated into the cornified envelope [Jonca *et al* 2023; Candi *et al*, 2005].

Compaction – Filaggrin (FLG)
Filaggrin protein is in charge of facilitating the aggregation and alignment of keratin filaments within corneocytes to ensure that these cells form a robust, cohesive layer on the surface of the skin. Mutations or deficiencies in Filaggrin have been linked to various skin disorders, including atopic dermatitis and ichthyosis vulgaris, highlighting its vital importance for healthy skin [Moosbrugger-Martinz *et al*, 2022].

Cohesion – Corneodesmosin (CDSN)
Corneodesmosin plays a fundamental role in maintaining the cohesion of the *stratum corneum* by ensuring strong and stable adhesion between corneocytes, the outermost cells of the epidermis. By keeping these cells firmly attached, corneodesmosin helps to create a continuous and unified skin barrier with a structural integrity that allows the skin surface to reflect light more evenly, contributing to a visibly radiant and smooth complexion [Jonca *et al*, 2010].

Together, involucrin, filaggrin, and corneodesmosin have key functions on the structure and cohesion of the *stratum corneum*, impacting how the skin reflects light. This is why we have named them as “Surface Optics proteins” or “Diamond Trio”.

miR-203 as Key Modulator of healthy epidermal layer



Micro-molecules also known as miRNAs (miR) are endogenous small non-coding RNAs that act as crucial gene regulators at posttranscriptional level by binding to the 3’ untranslated region (UTR) of target mRNAs. They are naturally present in the skin and mainly involved in skin phenotype which are related to skin barrier, photoaging, pigmentation, aging, senescence, acne and wound healing through either keratinocytes or fibroblasts, melanocytes or sebocytes [Li *et al*, 2021]. In the case of skin keratinocytes, the role of miRNA in epidermal barrier integrity has been identified (Lee *et al*, 2020). Moreover, loss of miRNA production exhibits taught and flaky skin, a sign of severe dehydration, demonstrating their role in the striking defects in epidermal differentiation [Yi *et al*, 2009].

At the epidermal architecture level, the disruption of miRNA processing in the skin resulted in abnormalities in epidermal stratification and miR-203 was the key miRNA implicated in driving this epidermal differentiation [Yi R *et al*, 2006 & 2008]. Likewise, miR-203 deletion resulted in a loss of all stratified epithelia [Schneider *et al*, 2012]. Finally, miR-203 was proposed as a therapeutic target for skin wound healing [Luo *et al*, 2021].

Within the skin, by promoting proper keratinocyte differentiation, the micro-molecule miR-203 helps in maintaining a healthy epidermal layer.

This process could be beneficial for creating a smooth, even skin texture that reflects light more effectively, thus enhancing the skin's radiance.

Cellular localization and functional demonstration of the epigenetic role of miR-203 in epidermal differentiation

This study investigates the epigenetic role of miR-203 in human epidermal differentiation by mapping its cellular localization in skin and functionally assessing its impact on key barrier-related gene expression in keratinocytes.



Figure 1 - Cellular localization of miR-203 in human skin biopsy using *in situ* hybridization (pink color).

miR-203 (also called miR-203a or mir-203a-3p) is an epigenetic small molecule viewable only in epidermis that has been described in the literature as a key regulator of keratinocyte differentiation and epidermal homeostasis, contributing to a finely balanced renewal of the epidermis [Golebiewski *et al*, 2023; Sonkoly *et al*, 2025]. miR-203 restricts the proliferative potential of basal keratinocytes, notably through repression of p63 [Yi *et al*, 2008] to improve keratinocyte differentiation.

FLG (filaggrin), IVL (involucrin), and CDSN (corneodesmosin) are essential structural components involved in the organization and cohesion of the *stratum corneum* through their roles in terminal keratinocyte differentiation, corneocyte maturation, and corneodesmosome mediated adhesion [Candi *et al*, 2005; Jonca *et al*, 2010; Moosbrugger Martinz *et al*, 2022].

In our study, we confirmed that miR-203 expression is strongly detected in epidermis suprabasal layers (Fig. 1). In addition, the artificial increase of miR-203 in keratinocyte cultures by transfection (Fig. 2A) triggers the increase of *IVL*, *FLG* and *CDSN* mRNA levels (Fig. 2B) confirming the strong implication of miR-203 in keratinocyte differentiation and consequently its involvement in skin radiance and hydration.

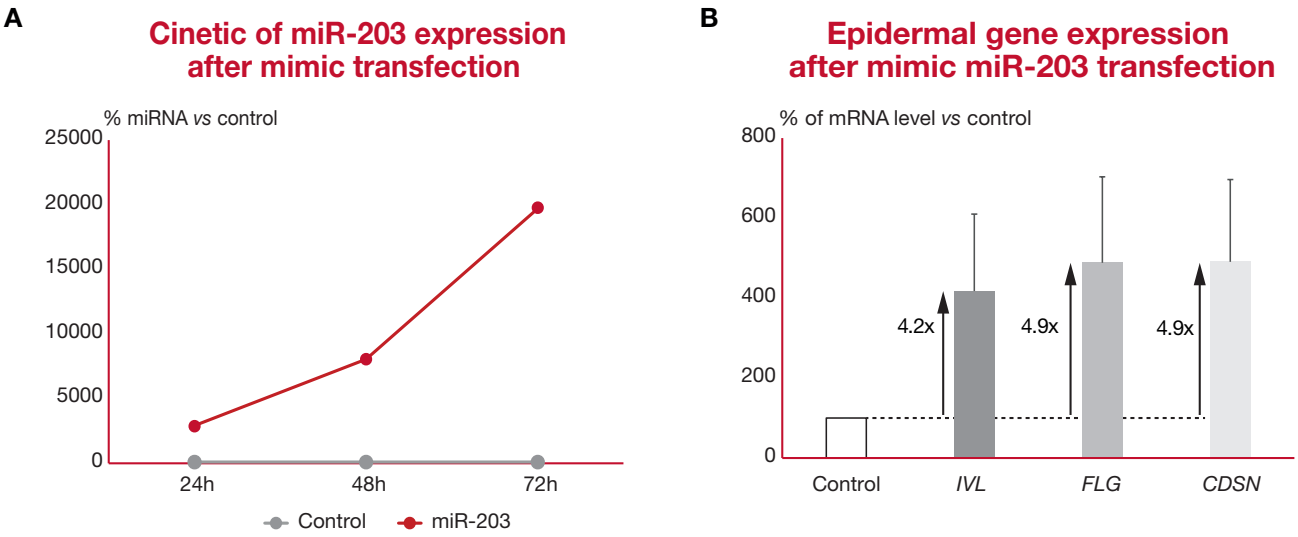


Figure 2 - Quantification of the miR-203 after keratinocyte transfection by mimic of miR-203 (**A**). Q-RT-PCR quantification of mRNA levels of 3 epidermal proteins 24h after transfection with the miR-203 mimic. Results are expressed in % against control mimic using a control of transfection (**B**).

MATERIALS&METHODS

miRNA visualization

Frozen skin explant was cut, sections mounted on superfrost plus slides and dried at -20°C. Sections were then fixed by 4% PFA at 4°C and dehydrated by successive 5 minutes incubations in ethanol at 50%; 70%; and twice at 100% before drying at room temperature. In situ hybridization was performed using the miRNAscope HD Assay Kit (ACD Biotechne), according to the manual instructions. Shortly, endogenous peroxidase was inactivated by incubation with Hydrogen Peroxide solution for 10 minutes at room temperature and then cells were permeabilized by Protease IV treatment for 30 minutes at room temperature before 2 PBS washings. Then, a specific probe against miR-203 was incubated on sections for 2 hours at 40°C before 2 buffer washings.

3 steps-signal amplification was performed, the last one corresponding to an alkaline phosphatase-labeled probe. Fast Red reactive was incubated for 10 minutes at room temperature to reveal the labelling and then washed. The slides were mounted, and color image acquisitions were performed using a Leica SP8 microscope.

Cell Transfection

Two strains of primary epidermal human keratinocytes were cultured until confluence . Transfections were performed using Lipofectamine RNAiMAX (Fisher scientific) following the manual instructions. Briefly, 13nmol of mimic miR-203 or mimic scramble control (Control) (Qiagen) was mixed with 9µl of lipofectamine RNAiMAX in 300µl EMEM and incubated for 5 minutes at room temperature for complex formation. Then, 250µl of the mix was applied on keratinocyte culture and incubated for 24h at 37°C. Cell culture medium was then changed and culture incubated for additional 24h or 48h.

RNA extraction and analysis

Long and short RNAs from keratinocyte cultures were extracted with NucleoSpin miRNA kit (Machery-Nagel), according to the manual instructions. Quantification and quality control of RNAs were performed by measurement of optical densities at 260 and 280 nm.

qRT-PCR for miR-203

Reverse transcriptions were realized on 0,1 µg of short RNA in 10 µl final mix using miRCURY LNA RT Kit (Invitrogen) and the C100 Touch Thermocycleur (BioRad) for each condition.

Quantitative PCRs were performed on 3 µl of RT mix, diluted at 1/60 or 1/600, according to the culture, in 10 µl final reaction, using specific primers for miR-203, according to manual instruction (miRCURY LNA SYBR Green PCR Kit, Qiagen) in a LC480 thermocycler (Roche).

Each strain analysis was performed in duplicate, and the mean used to calculate miRNA expression compared to a housekeeping gene (SNORD48, small nucleolar RNA, C/D box 48) analyzed in duplicate.

qRT-PCR for epidermal gene expression

Reverse transcriptions were realized on 0,1 µg of long RNA in 20 µl final mix using SuperScript IV VILO Master Mix Kit (Invitrogen) and the C100 Touch Thermocycleur (BioRad) for each condition.

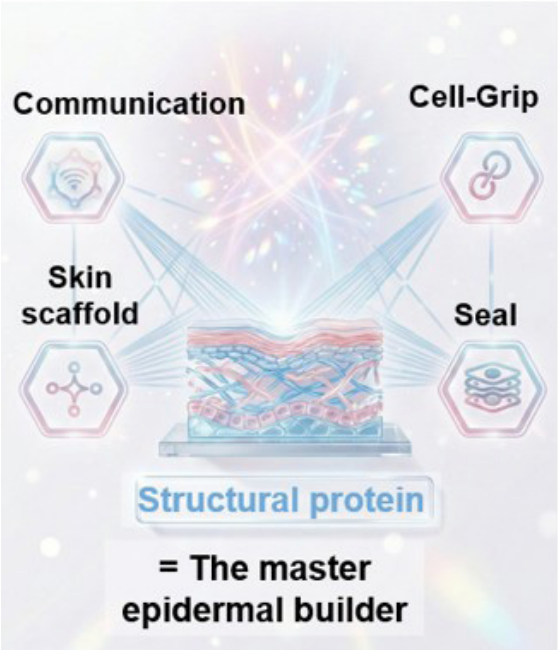
Quantitative PCRs were performed on RT mix in 10 µl final reaction, according to manual instruction (LC 480 SYBR Green I Master, Roche) in a LC480 thermocycler (Roche).

Each strain analysis was performed in duplicate, and the mean used to calculate gene expression. Results were normalized compared to the mean of housekeeping genes analyzed in duplicate: *ACTB* and *SDHA* (coding for Actin B and succinate dehydrogenase subunit A, respectively) for *CDSN* and *FLG* analysis or only *SDHA* for *IVL* analysis.

Results

Results are expressed as a percentage referring to the control condition. For kinetic, miR-203 levels, analyzed by qRT PCR, are presented as relative expression values normalized to the corresponding reference. For epidermal gene expression, qRT PCR results are expressed as mean ± SEM (n=2).

Perfection is structure



In epidermis, cell–cell adhesion complexes, including adherens junctions (AJs), tight junctions (TJs), gap junctions (GJs), and desmosomes, are essential mechanical interactions for maintaining epithelial homeostasis. Their disruptions can disturb the epidermal barrier, resulting in skin disorders such as psoriasis, dermatitis, and keratosis [Tatari *et al*, 2014].

TJs are composed of **seal proteins** such as claudins and occludins, responsible for forming a tight barrier between cells, preventing the passage of substances between them. They are in charge of regulating permeability and protecting against pathogens, they create a selective barrier. Claudin 1 & 4, TJP1&2 and Occludin belong to these complexes.

GJs are specialized intercellular connections that enable direct **communication** between adjacent cells to coordinate cellular activities. The mains proteins are connexins that facilitate rapid communication between cells that is vital for maintaining cellular functions and responding to changes in the environment [Martin *et al*, 2014]. Connexin 31 (GJB3) belongs to these complexes.

Desmosomes anchor keratinocytes together, helping to maintain the structural integrity of the epidermis by distributing mechanical stress across the tissue, making the epidermis more resilient to physical forces. They contribute to the overall barrier function of the skin, helping to prevent the entry of pathogens and water loss [Perl *et al*, 2024]. Desmoplakin, Desmoglein and Desmocollin belong to these complexes. Their function contributes to strengthening intercellular cohesion, a key biological foundation of what we describe as the "**cell-grip**" effect.

AJs are composed of cadherins ensuring cell adhesion and the transmission of mechanical forces between cells, contributing to the cohesion and structure of the epidermal **scaffold**. E-cadherin belongs to these complexes [Troyanovsky *et al*, 2023].

Envoplakin and Periplakin are the smallest members of the plakin family, that play a structural role in **scaffolding** and assembling the cornified envelope during epidermal differentiation [Boczonadi *et al*, 2016]. The loss of these plakins is associated with several skin pathologies, including ichthyosis, a condition characterized by dry, scaly skin associated with defective epidermal barrier [Sevilla *et al*, 2007].

The complexity of epidermal cohesion is beneficial for the skin, enabling it to adapt to various environmental stimuli and maintain its integrity. Moreover, the diversity of proteins and lipids in the extracellular matrix reinforces the skin barrier, protecting against infection, dehydration and chemical aggression. Such flexibility and resilience are essential for skin health and functionality.

Traditional and organic *Rosa x damascena*

Natural extract of Rosa x damascena flowers from sustainable and ethical floral culture

Description of the plant

Family: Rosaceae

Botanical name: *Rosa × damascena Mill*

Vernacular name: four-seasons rose, damask rose, Portland rose, York-and-Lancaster rose (english), rosier de Damas (French)

Synonym: *Rosa gallica var. damascena Voss*

Description & Distribution of the plant: *Rosa × damascena* is a rose hybrid, derived from *Rosa gallica* and *Rosa moschata* or possibly *Rosa fedtschenkoana*. It is a deciduous shrub growing to 2.2 m high, the stems densely armed with stout, curved prickles and stiff bristles. The leaves are pinnate, with 5-7 leaflets. The relatively small flowers grow in groups and are light to moderate pink to light red colored. *Rosa × damascena* is a cultivated flower that is not found growing wild. It originated from west Asia and was brought to Europe for cultivation.

Habitat: it grows in temperate regions, often at altitudes up to 1,200 meters, in open, sunny areas. The plant prefers well-drained, calcareous or loamy soils and is sensitive to excessive winter moisture.

Spectral analysis of phenolic compounds using confocal microscopy in sections of rose flower peduncles.



Figure 3 - histological-localization of polyphenols in cross section of rose peduncle, using Neu's reagent. Spectral study using confocal microscope. Kaempferol-glycosides (green/yellow), gallic acid- and ellagic acid-based tannins (blue) and quercetin-glycosides (red/pink).

The image (Fig. 3) reveals the localization of 3 polyphenols in the rosa peduncle. The green/yellow fluorescence is related to the kaempferol-glycosides with an absorption at 488 nm and emission between 510 and 550 nm . Moreover, gallic acid- and ellagic acid-based tannins (tannins) are visible by their blue fluorescence due to their absorption band centered at 405 nm and their optimal emission between 450 and 480 nm. Then, the red/pink fluorescence reveals the quercetin-glycosides due to their absorption at 552 nm and emission between 560 and 640 nm.

MATERIALS&METHODS

Flower peduncle cross-sections (100 µm) were obtained using a blade hand microtome and then dipped in Neu's reagent then in PEG (polyethylene glycol) and glycerol/water (50:50) solution. Briefly, Neu's reagent reacts with phenolic groups emitting a specific color epifluorescence under UV light regarding different classes of polyphenols [Neu *et al*, 1956].

Spectral study was conducted using a confocal microscope (Leica Microsystems SAS, Nanterre, France) to characterize the fluorescence absorption and emission properties of three polyphenol families: gallic acid- and ellagic acid- based tannins, kaempferol-glycosides and quercetin-glycosides.



Origin: Morocco

Part used: Flower buds

Sourcing and harvest: Damask rose is cultivated in Morocco, in the Marrakech-Safi region, according to organic farming practices and to sustainable sourcing standards.

The sourcing approach of Floragenist Rose supports the goals of BASF's Responsibly Active program for its cosmetic bioactives portfolio.

For more information about the sourcing, please refer to Botanical sourcing information available upon request.

Traditional and modern uses

Rosa x damascena is used as an ornamental plant in parks, gardens, and houses. Damask roses are used in cooking as a flavoring ingredient or spice. Rose water and powdered roses are used in Persian, Indian, and Middle Eastern cuisine. Whole flowers, or petals, are also used in herbal tea. The most popular use, however, is in the flavoring of desserts such as ice cream, jam, Turkish delights, rice pudding, yogurt…

In Iranian traditional Medicines, the decoction of flowers was used for treatment of chest and abdominal pains, menstrual bleeding, and digestive ailments. It has been famous as cardiotonic agent for strengthening the heart.

Essential oil was used for the treatment of different ailments. Rose water was traditionally used as antiseptic agent for eye, mouth disinfecting and as antispasmodic agent for alleviating abdominal pains, and bronchial and chest congestion.

AquaGenesis technology: use of coconut water as natural solvent for enriched plant extracts

The use of Cocos nucifera fruit for health and Beauty

The fruit of the coconut palm (Cocos nucifera L.) has probably been used by humans for centuries. Nearly one-third of the world's population depends, to some extent, on coconuts for their food and economy. In these cultures, coconuts have a long and respected history [Obidoa *et al*, 2009]. Coconut is classified as a "functional food" because it offers many health benefits, beyond its nutritional content. The edible endosperm is used either as a solid white kernel or as a clear nutrient liquid (coconut water) [Prades *et al*, 2012a; Prades *et al*, 2012b]. In particular, the coconut water is known to possess a range of health-promoting nutrients, and it is commonly used for food, pharmaceutical or cosmetic applications [Kumar *et al*, 2021].

New use of Cocos nucifera fruit for Bio-based technologies

Aqueous extraction, commonly used in the manufacture of plant-based cosmetics, is considered as safe and environmentally friendly method. Research by BASF and the University of Avignon and the Vaucluse region (in France) has shown that coconut water can be used as a natural solvent to improve plant extraction, thereby enhancing the phytochemical content [Makerri *et al*, 2018]. This patented technology, offers new benefits for the cosmetics industry.

The coconut water is sourced from organically cultivated dwarf coconut trees in Tamil Nadu, India, and is hand-harvested year-round. Extracted from young green coconuts aged 6–9 months, the entire fruit is valorized to minimize waste.

What brings this patented AquaGenesis extraction technology?

AquaGenesis is an extraction technology that uses natural coconut water as solvent to extract rose flowers. Compared with aqueous extraction, it enriches the phytochemical compounds of the extract.

Controlled particle size and porosity of botanical raw material (= matrix) influence extraction efficiency while preserving process robustness. Temperature control accelerates diffusion and solubility without compromising molecular integrity [Chemat F, Rombaut N *et al*, 2017].

An extraction solvent is characterized by different physico-chemical parameters and the key relevant ones for the use of coconut water as extraction solvent are the following: pH, osmolarity, ionic strength and surface tension [Yong *et al*, 2009].

Since surface tension belongs to the physical parameters that is directly linked to the enhanced penetration of solvents into plant tissues, facilitating the release of phytochemical compounds [Samant *et al*, 2025], we compared coconut water and water as solvents based on this property.

Indeed, the surface tension of water at 20°C is well described in literature (≈ 72 $mN \cdot m^{-1}$), data that was confirmed by internal measures. The presence of sugars and electrolytes reduces this value of 33% for coconut water at the same temperature. With a decreased surface tension of the solvent referring to water, the wetting of the matrix (grinded rose flowers) is enhanced allowing better solvent access. This results in a more efficient diffusion of targeted phytochemical compounds of rose flower with the coconut water.

Benefits of AquaGenesis technology process on rose extraction

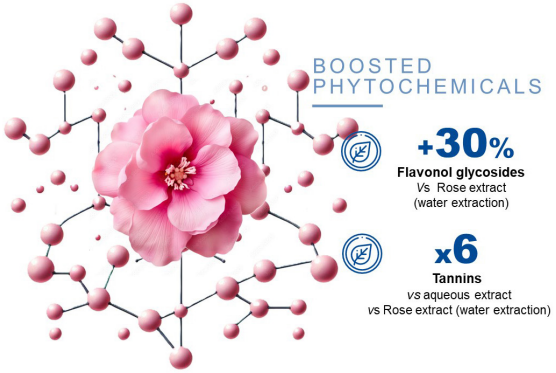


Figure 4 - Enriched phytochemicals of rose extract with AquaGenesis technology process *vs* water.

We have performed a phytochemical comparison of rose extracts obtained using coconut water or water. The results revealed an enriched phytochemical profile with coconut water: 6x in tannins, and +30% in flavonol glycosides *vs* water (Fig. 4).

Additionally, *in vitro* investigations on CDSN protein synthesis in human keratinocytes demonstrated that coconut water-rose extract outperforms water-rose extract, boosting significantly by 2x (*vs* water extract, $p<0.01$) the level of the surface protein corneodesmosin known for strengthening **cell cohesion in the *stratum corneum***.

AquaGenesis technology enables a more complete phytochemical extraction, producing an enriched rose extract.

MATERIALS&METHODS

Tested products

Freeze-dried coconut-water-rose Extract (batch CS220321.01; INCI: Cocos Nucifera (Coconut) Fruit Extract (and) Rosa Damascena Flower Extract) and Freeze-dried water-rose extract (batch EXT220921.01) were prepared with similar content of rose raw material.

Control coconut water solvent without rose raw material (batch EXT220524.01) was used to validate that tannins and flavonol glycosides came only from the flowers (data not shown). Moreover, the control coconut water was used for surface tension investigations.

Tannins and flavonol glycosides measurement

Gallic acid- and ellagic acid-based tannins (tannins) were quantified by the Folin–Ciocalteu (F–C) assay, commonly used for the quantification of total phenolic content in biological samples, based on the reaction of phenolic compounds with a colorimetric reagent. The principle of the assay relies on the transfer of electrons in alkaline medium from phenolic compounds to phosphomolybdic/phosphotungstic acid complexes to form blue complexes that are determined spectroscopically at approximately 750 nm [Ainsworth *et al*, 2007].

Flavonol glycosides were quantified by high-performance liquid chromatography coupled to a photodiode array detector (UHPLC–PDA) using an Ultimate 3000 system (Thermo Scientific Dionex). The different molecules presented in figure 6 were quantified to obtain the total flavonol glycoside content as the sum of quercetin-3-O-galactoside, kaempferol-3-O-glucoside, and their respective equivalents.

CDSN protein

Normal human keratinocytes were incubated 3 days at 37°C and 5% CO_2 in MCDB153 medium in the presence of Floragenist Rose or without any product (untreated control). Lack of cytotoxicity of the tested products was checked by MTT test (data not shown). The cellular corneodesmosin was measured using a specific commercial ELISA method (Sigma).

Results and statistics

Results are expressed as a percentage referring to water-rose extract condition and presented as a Mean ± SEM (n=6). The statistics are based on Welsch t test with a threshold of significance set at 5% ($p<0.05$).

Processing

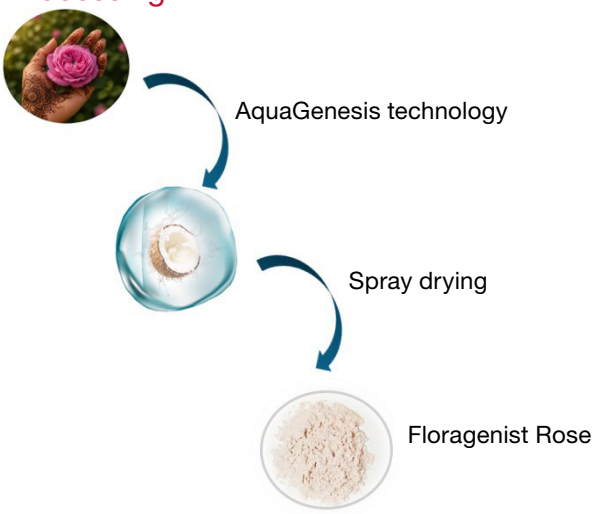


Figure 5 - Processing.

Floragenist Rose is obtained through patented AquaGenesis extraction process that uses coconut water as the solvent, applied to dried and crushed rose flower buds (Fig. 5). After removing insoluble matter, the extract is concentrated and transformed into a spray-dried, preservative-free powder with maltodextrin. Floragenist Rose is characterized by its content in flavonol glycosides (≥ 0.1%) and tannins.

Composition

The typical composition of coconut-water-rose extract is detailed in Table 1. Among the principal constituents extracted from the rose flower, flavonol glycosides are identified by UHPLC-PDA (Fig. 6), and quantified as equivalent of kaempferol 3-O-glucoside and quercetin 3-O-galactoside. Gallic acid- and ellagic acid-based tannins are evaluated by the spectrophotometric method, employing the Folin-Ciocalteu reagent and quantified in tannic acid equivalent.

Carbohydrates are quantified by total sugars analysis using the colorimetric phenol method. In addition, protides (amino acids, peptides & proteins) are determined by measuring total nitrogen via the Kjeldhal method, applying a conversion factor of 6.25. Furthermore, minerals and salts are measured through ash measurement by calcination.

Components	Range %
Carbohydrates	47.6 - 59.8
Minerals and salts	6.4 - 11.2
Protides (amino-acid and proteins)	4.0 - 8.6
Gallic acid- and ellagic acid-based tannins	2.0 - 4.9
Flavonols glycosides	1.0 - 1.7

Table 1 - Typical phytochemical composition of coconut-water-rose extract (Cocos Nucifera (Coconut) Fruit Extract (and) Rosa Damascena Flower Extract) in Floragenist (weight % on extract, based on 5 representative batches).

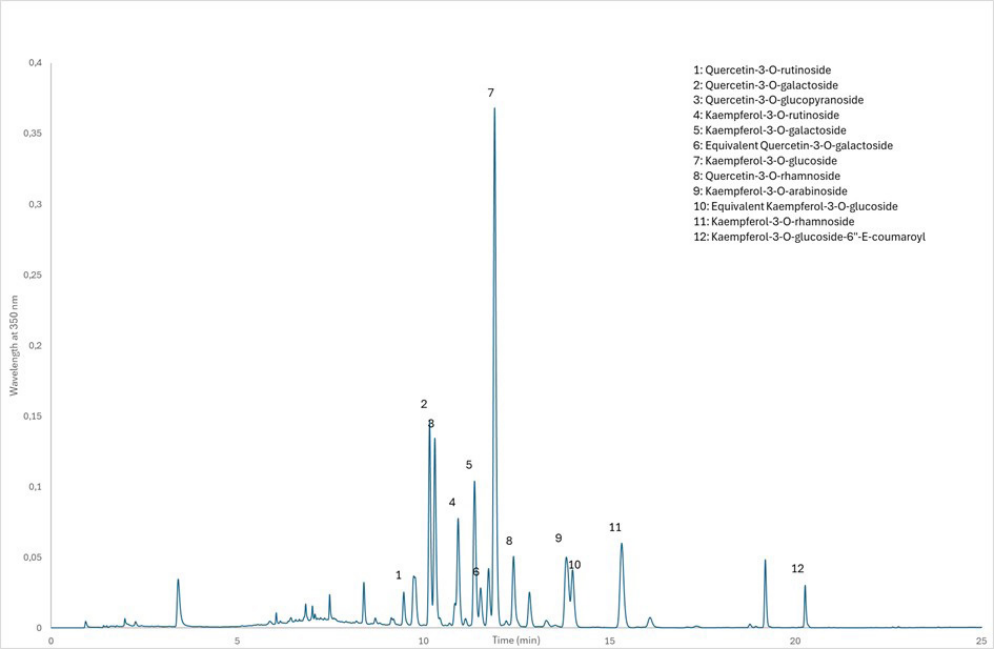


Figure 6 - Example of chromatogram profile obtained by UHPLC-PDA for Floragenist Rose.

Safety / tolerability of the product

Floragenist Rose was evaluated to ensure its safety in the recommended conditions of use.

Description

Floragenist Rose is an extract of organic *Rosa damascena* flower, obtained through the patented AquaGenesis technology, titrated in flavonol glycosides, available as a preservative-free powder.

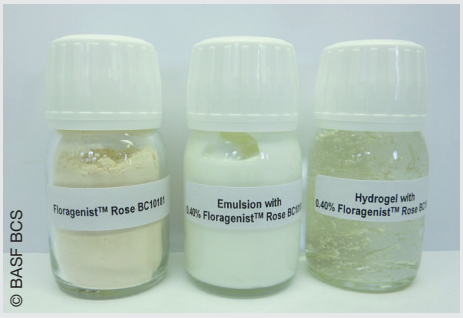
INCI name: Maltodextrin (and) Cocos Nucifera (Coconut) Fruit Extract (and) Rosa Damascena Flower Extract

Naturalness content (according to ISO 16128): 100% from natural origin

Recommended dose of use: 0.4%

China IECIC listed

Approved by ECOCERT RAW MATERIAL COSMOS APPROVED



Example: illustration of commercial samples of Floragenist Rose with a hydrogel and emulsion containing 0.4% of Floragenist Rose.



Demonstrated efficacy

Intricated processes drive skin surface perfection – including keratinocyte differentiation and cohesion under epigenetic control that help to lay the foundation for visible radiance and even tone.

Floragenist Rose was evaluated through *in vitro* studies on keratinocytes to demonstrate its capacity to boost an epigenetic micro-molecule miR-203. This micro-molecule contributes to skin differentiation and cohesion by regulating oxidative and inflammatory stress induced by environmental damage.

In vitro, the properties of Floragenist Rose were evaluated:

- on the stimulation of cell differentiation and cohesion by stimulating miR-203, filaggrin, involucrin (mRNA) and corneodesmosin (mRNA and protein) on keratinocytes,
- on the stimulation of the corneodesmosin protein on keratinocyte (*in vitro*),
- on the stimulation of expression of genes involved in keratinocyte cell–cell adhesion complexes (seal proteins, adherens junctions, gap junction and desmosomes),
- on the stimulation of radical scavenging and photoprotection effect on keratinocytes against UVA,
- on the decrease of the release of pro-inflammatory cytokines in UVB-stressed keratinocytes.

In vivo, Floragenist Rose was evaluated on different categories of parameters contributing to **skin perfection** improvement: redness and luminosity parameter, **homogeneity and healthy skin** with expert grading and perceived luminosity with a self-perception questionnaire.

EFFICACY

Stimulation of miR-203, Filaggrin, Involucrin and Corneodesmosin on keratinocytes

In vitro

OBJECTIVE

As demonstrated in our previous study miR-203 is strongly expressed in epidermis suprabasal layers, and its expression triggers the increase of involucrin (*IVL*), filaggrin (*FLG*) and corneodesmosin (*CDSN*). mRNA levels.
The aim of the present study was to evaluate the effect of Floragenist Rose on the miR-203 expression by keratinocytes, and as a consequence on the expression of the differentiation and cell cohesion gene *IVL, FLG* and *CDSN*.

RESULTS & DISCUSSION

Visualization of the miR-203 and involucrin changes in keratinocytes cultures

To investigate the correlation between miR-203 and differentiation, we used Fluorescence *in Situ* Hybridization (FISH) technology to visualize miR-203 (in green) and *IVL* mRNA (in yellow) at the same time in keratinocyte monolayers treated with or without Floragenist Rose at 0.04%. We showed that the increase of miR-203 was correlated with a simultaneous visual increase of *IVL* mRNA at similar location in cells (Fig. 7).

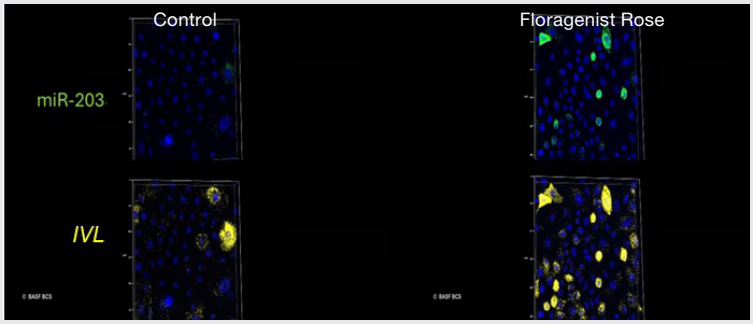


Figure 7 - Visualization of miR-203 (green fluorescence) and *IVL* mRNA (yellow fluorescence) in keratinocyte culture treated by Floragenist Rose or not (Control). Nuclei were stained with DAPI in blue. Illustrative picture.

Increase of miR-203

Human keratinocytes were treated with Floragenist Rose, total RNA was then isolated and subjected to RT-qPCR for the expression of the mature miR-203. This miR-203 was significantly up-regulated in keratinocytes treated with Floragenist Rose (Fig. 8). miR-203 was increased by 3.3-fold change at 0.09% (± SEM 49.7%, p<0.001 *vs* UC normalized at 100%, SEM 0%).

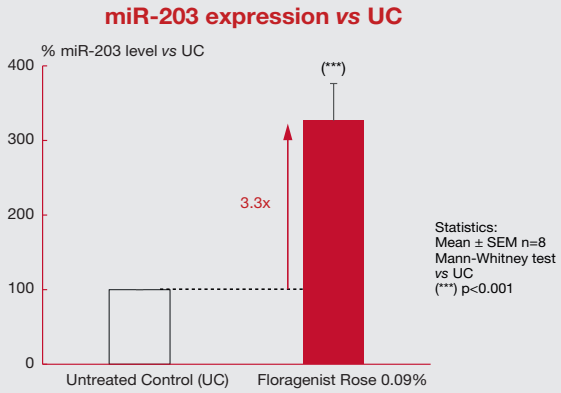


Figure 8 - Expression of miR-203 in human keratinocytes treated with or without Floragenist rose 0.09%.

Stimulation of the mRNA expression levels of the *IVL*, *FLG* and *CDSN* genes

To study the effect of Floragenist Rose on keratinocyte differentiation, human keratinocytes were treated with 0.09% Floragenist Rose. Total RNA was then isolated and subjected to reverse transcription-qPCR using the primers for *IVL*, *FLG* and *CDSN*.

As expected from the stimulatory effect on miR-203 level, the result showed a significant increase of the mRNA expression levels of the 3 genes after treatment. (Fig. 9). *IVL* gene expression level was increased by 3.4-fold change (± SEM 144%, p<0.001 *vs* UC normalized at 100%, SEM 0%), *FLG* by 2.4-fold change (± SEM 64%, p<0.001 *vs* UC normalized at 100%, SEM 0%), and *CDSN* by 5-fold change (± SEM 119%, p<0.001 *vs* UC normalized at 100%, SEM 0%).

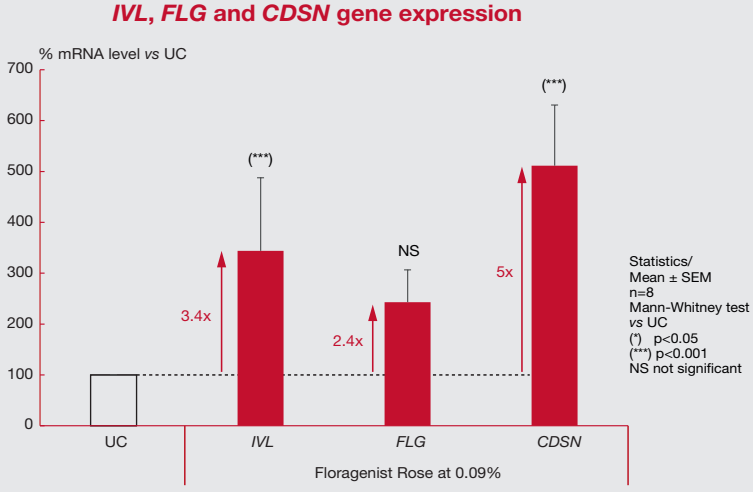


Figure 9 - Evaluation of *IVL*, *FLG* and *CDSN* mRNA stimulation by Floragenist Rose at 0.09% stimulation of expression.

In vitro

CONCLUSION

In vitro on keratinocytes, Floragenist Rose stimulates the expression of miR-203 triggering an increase of the gene expression of *IVL*, *FLG* and *CDSN*. It thus may contribute to improve epidermal differentiation and *stratum corneum* cohesion, leading to a smoother and radiant skin.

MATERIALS&METHODS

Study number

Study EMB161102-2022-709-BM

Tested products

Freeze-dried coconut-water-rose Extract (batch CS220321.01; INCI name: Cocos Nucifera (Coconut) Fruit Extract (and) Rosa Damascena Flower Extract) was tested diluted in culture medium at concentrations corresponding to Floragenist Rose at 0.09% for qRT-PCR & 0.04% for visualization.

RNA extraction and analysis

Long and short RNAs from keratinocyte cultures were extracted with NucleoSpin miRNA kit (Macherey-Nagel), according to the manual instructions. Quantification and quality control of RNAs were performed by measurement of optical densities at 260 and 280 nm.

qRT-PCR for miR-203

Reverse transcriptions were realized on 0.1 µg of short RNA in 10 µl final mix using miRCURY LNA RT Kit (Invitrogen) and the C100 Touch Thermocycleur (BioRad) for each condition.

Quantitative PCRs were performed on 3 µl of RT mix, diluted at 1/100 in 10 µl final reaction, using specific primers for miR-203 according to manual instruction (miRCURY LNA SYBR Green PCR Kit, Qiagen) in a LC480 thermocycler (Roche).

Each analysis (4) was performed in duplicate and the mean used to calculate gene expression compared to 2 housekeeping genes (SNORD48 & RNU6-1 producing small nucleolar RNA, C/D box 48 & small nuclear RNA U6).

qRT-PCR for epidermal gene expression

Reverse transcriptions were realized on 0.1 µg of long RNA in 20 µl final mix using SuperScript IV VILO Master Mix Kit (Invitrogen) and the C100 Touch Thermocycleur (BioRad) for each condition.

Quantitative PCRs were performed on 2 µl of RT mix, diluted at 1/2 for *IVL*, *CDSN*, or at 1/50 for *FLG*, in 10 µl final reaction, according to manual instruction (LC 480 SYBR Green I Master, Roche) in a LC480 thermocycler (Roche).

Each strain analysis was performed in duplicate and the mean used to calculate gene expression compared to the mean of 2 housekeeping genes analyzed in duplicate: *ACTB* and *SDHA* (coding for Actin B and succinate dehydrogenase subunit A, respectively) for *IVL, CDSN* or *ALAS1* and *PPIA* (coding for 5'-aminolevulinate synthase 1 and peptidylpropyl isomerase A, respectively) for *FLG* analysis.

miRNA visualization (FISH)

After human keratinocyte culture and fixation, slides were washed twice in PBS and then dehydrated by successive 5 minutes incubations in ethanol at 50%; 70%; and twice at 100% before drying at room temperature.

In situ hybridization were performed using the RNAscope Plus smRNA-RNA HD Reagent Kit (ACD Biotechne), according to the manual instructions. Shortly, endogenous peroxidase was inactivated by incubation with Hydrogen Peroxide solution for 10 minutes at room temperature and then cells were permeabilized by Protease III treatment for 30 minutes at room temperature before 2 PBS washings. Then specific probes against miR-203 or *IVL* mRNA were mixed and incubated on cells for 2 hours at 40°C before 2 washings with washing buffer.

Signal amplification was performed for each probe using a 3 steps revelation, the last one corresponding to a specific fluorochrome (TSA Vivid Florochrome 520 or TSA Vivid Florochrome 570 for miR-203 or *IVL* revelation, respectively). After washing, cell nuclei were labelled by DAPI and then slides mounted. Fluorescent image captures were performed using a Leica SP8 microscope.

Results and statistics

Results are expressed as a percentage referring to UC condition and presented as a Mean ± SEM (n=8). The statistics are based on Mann-Whitney Rank Sum test with a threshold of significance set at 5% ($p<0.05$).

EFFICACY

Stimulation of the corneodesmosin protein on keratinocyte

in vitro

OBJECTIVE

To identify the molecules involved in the activity that stimulates keratinocyte differentiation, Floragenist Rose was fractionated using phytochemical methods, and each fraction was evaluated for its effect on corneodesmosin in keratinocytes.

RESULTS & DISCUSSION

Stimulation of corneodesmosin protein by Floragenist Rose extract

In addition to its stimulating effect on CDSN mRNA level, Floragenist rose was shown to increase the CDSN protein level by 3 – folds ($p<0.01$) at 0.045% (Fig. 10).

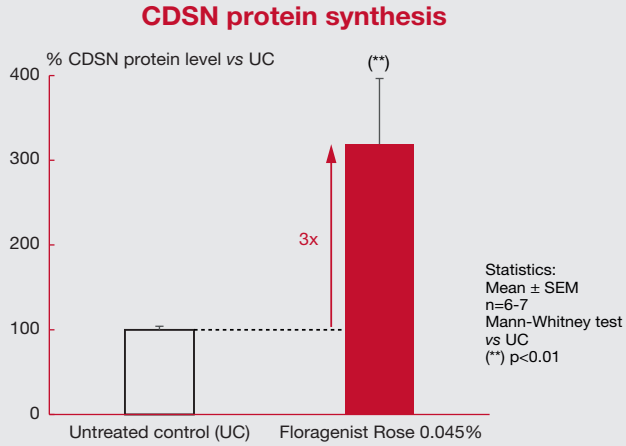


Figure 10 - Quantification of CDSN protein level in human keratinocytes treated with 0.045% of Floragenist Rose.

Stimulation of corneodesmosin protein by fractions of Floragenist Rose

Eleven fractions were obtained by polarity fractionation and tested at isodose on keratinocytes, assessing CDSN protein levels. Isodose conditions were applied, meaning that each fraction was tested at a concentration proportional to its relative abundance in the original rose extract.

Among the 11 fractions, 4 (F04, F05, F06 and F09) showed significant stimulation of CDSN protein vs the untreated control (set at 100%, SEM ± 3%): by respectively 75% (± SEM 15%, $p<0.001$), 88% (± SEM 4.7%, $p<0.001$), 101% (± SEM 5.7%, $p<0.001$ *vs* UC) and 96% (± SEM 9.3%, $p<0.01$) (Fig.11).

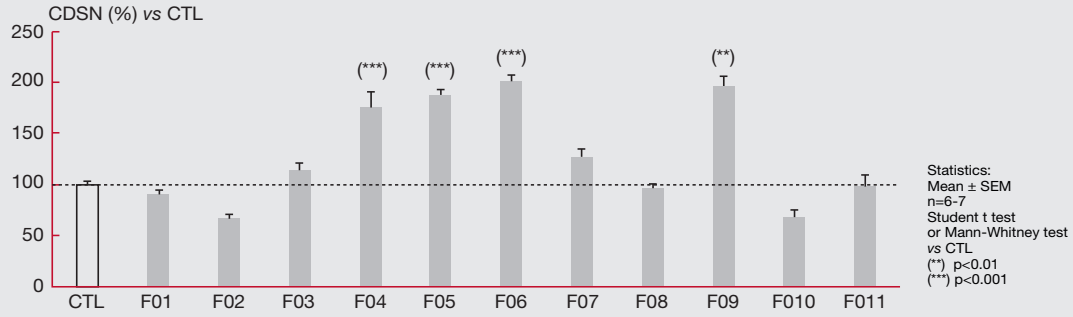


Figure 11 - CDSN protein stimulation by fractions of Floragenist rose obtained by polarity fractionation.

Actually, 3 out of the 4 active fractions (F04, F05, F06) contain Flavonol glycosides and the tannin molecules including tellimagrandin which was already described as effective on CDSN protein stimulation [Pannakal *et al*, 2023]. However, the 9th fraction which was a "tannin free" fraction (data not shown) showed similar activity on CDSN protein stimulation. Interestingly, this 9th fraction presents potentially molecules with protective effects described on human epidermal keratinocytes exposed to UV-induced cellular damage. For instance, Rutin, a flavonol glycoside also known as rutoside (Quercetin-3-O-rutinoside), has been shown to safeguard epidermal integrity under photo-induced stress [Gęgotek *et al*, 2019].

CONCLUSION

Fractions 04, 05, 06 and 09 of Floragenist Rose demonstrated *in vitro* properties on keratinocyte differentiation.

MATERIALS&METHODS

Study number
EMB161102-2022-710-CC

Tested products
Freeze-dried coconut-water-rose extract (batch CS220321.01; INCI name: Cocos Nucifera (Coconut) Fruit Extract (and) Rosa Damascena Flower Extract) was tested diluted in culture medium at concentrations corresponding to Floragenist Rose at 0.045% for CDSN protein quantification. Moreover, 11 fractions (F01 to F11) of the coconut-water-rose extract matter from Floragenist Rose, obtained by centrifugal partition chromatography (CPC), were tested at isodose of the 100% active matter of Floragenist Rose. Based on the percentage of each fraction present in the rose extract, the isodose was calculated for each fraction.

Cell culture
Normal human keratinocytes (NHEK), obtained from healthy donors, were seeded in 24-well plates at a rate of 35,000 cells/cm² and cultured in MCDB153 supplemented with 2% FCS, 0.5% antibiotic 96h at 37°C, under 5% CO_2. Then cells were incubated during 72h with the products or without any product (CTL) in MCDB153 supplemented with 0.5% antibiotic.

MTT assay
The lack of cytotoxicity of the treatments was checked thanks to the Thiazolyl Blue Tetrazolium Bromide (MTT) reagent. Briefly, after washing, cells were incubated with MTT at 0.25 mg/mL for 2h30 at 37°C. Then, the cells were washed and incubated with 1mL of Dimethylsulfoxide (DMSO). The absorbance was read at 540 nm.

CDSN protein measurement
Normal human keratinocytes were incubated 3 days at 37°C and 5% CO_2 in MCDB153 medium in the presence of Floragenist Rose or without any product (untreated control). Lack of cytotoxicity of the tested products was checked by MTT test (data not shown). The cellular corneodesmosin was measured using a specific commercial ELISA method (Sigma).

Results and statistics
Results are expressed as a percentage referring to Untreated Control condition and presented as a Mean ± SEM (n=6-7). The statistics are based on Mann-Whitney Rank Sum test with a threshold of significance set at 5% ($p<0.05$).

EFFICACY

Stimulation of expression of genes involved in cell–cell adhesion complexes

In vitro

OBJECTIVE

The structure of the epidermis relies on a network of cell–cell adhesion complexes—including tight junctions ("Seal proteins" (Claudin 1 & 4, TJP1&2 and Occludin)), adherens junctions ("Scaffolding" (E-Cadherin, Envoplakin and Periplakin)), gap junctions ("Communication" (Connexin 31)), and desmosomes ("Cell-grip" effect: (Desmoplakin, Desmoglein and Desmocollin))—which collectively maintain skin integrity and homeostasis. Each complex, composed of specific proteins, plays a distinct role in regulating permeability, mechanical stability, and cellular communication. Disruptions in these systems can lead to various skin disorders [Tatari *et al*, 2014], highlighting the importance of their coordinated function to help to reinforce the skin's protective capacity.
The aim of this study was to evaluate the effect of Floragenist Rose on cell–cell cohesion genes in keratinocyte cultures using a specific thematic PCR targeting the expression of 15 genes involved in epithelial cell components of the junctional complex.

RESULTS & DISCUSSION

A direct comparison of the relative expression levels in Floragenist Rose condition *vs* control cell cultures revealed that the most highly expressed genes included network of cell–cell adhesion complex genes.
Floragenist rose treatment showed a clear stimulation of 12 gene expressions with more than fold-change > 2 (Fig. 12) with 2 doses of Floragenist Rose. The highest expressed genes were Periplakin (5.3x) Desmocollin (4.4x) and Claudin 4 (9.6x) respectivelly related to adherent junctions, desmosomes and tight junctions.

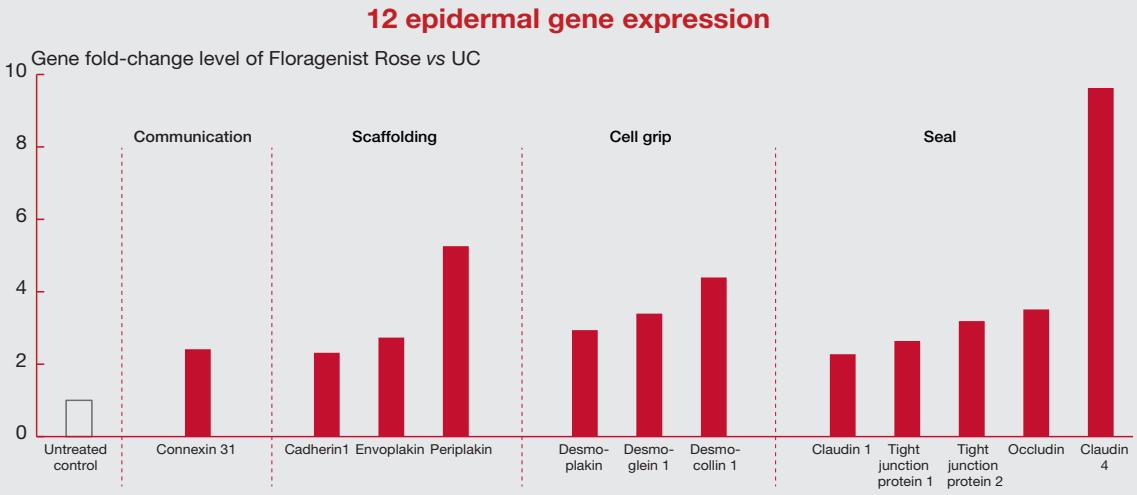


Figure 12 - Change in mRNA level of different genes involved in epidermal structure in keratinocytes treated with Floragenist Rose.

CONCLUSION

In vitro, Floragenist Rose stimulates the expression level of 12 genes encoding proteins involved in keratinocyte cell-cell adhesion complexes, and thus may contribute to improve the epidermal architecture and cohesion for a smoother and radiant skin.

MATERIALS&METHODS

Study number
EMB161102-2023-822-BM

Tested products
Freeze-dried coconut-water-rose Extract (batch CS220321.01; INCI name: Cocos Nucifera (Coconut) Fruit Extract (and) Rosa Damascena Flower Extract) was tested diluted in culture medium at concentrations corresponding to Floragenist Rose at 0.045 or 0.09% for gene expression quantification.

RNA extraction and analysis
Total RNAs from keratinocyte cultures were extracted with NucleoSpin miRNA kit (Macherey-Nagel), according to the manual instructions. Quantification and quality control of RNAs were performed by measurement of optical densities at 260 and 280 nm.

Reverse transcriptions were realized on 0,2 µg of RNA in 40 µl final mix using SuperScript IV VILO Master Mix Kit (Invitrogen) and the C100 Touch Thermocycleur (BioRad) for each condition.
For almost all gene analyses, quantitative PCRs were performed on 2 µl of RT mix, diluted at 1/2 in 10 µl final reaction, according to manual instruction (LC 480 SYBR Green I Master, Roche) in a LC480 thermocycler (Roche). For *CLDN1* & *DSC1* gene expression, quantitative PCRs were performed on 2 µl of RT mix, diluted at 1/20 in 10 µl final reaction,

Results and statistics
Each analysis was performed in duplicate and the mean used to calculate gene expression compared to a housekeeping gene (*SDHA*: succinate dehydrogenase subunit A) analyzed in the same condition compared to the target gene. Due to the weak number of cultures studied (2), no statistical analysis was performed. Fold change > 2 corresponds to a high stimulation level.

EFFICACY

From radical scavenging to cellular photoprotection

In vitro

OBJECTIVE

Consistent application of antioxidants could contribute to improving the skin's look by encouraging a more vibrant and healthier complexion, commonly known as a "glow" [Goodman *et al*, 2024]. Reactive oxygen species (ROS) and lipid peroxides through MDA production are crucial factors in the onset and advancement of accelerated skin aging, particularly when there is extended oxidative stress [Schwartz *et al*, 2016]. These reactive species can be generated by various environmental factors, among which skin exposure to ultraviolet radiation (UVR) also results in the formation of ROS [Khan *et al*, 2018].
The aim of the study was to assess the free radical scavenging and antioxidant properties of Floragenist Rose to support skin defense against UVA.

RESULTS & DISCUSSION

Antioxidant protection (*in tubo*)

The free radical scavenging properties of Floragenist Rose were evaluated by the DPPH° (diphenylpicryl-hydrazyl) assay. Floragenist Rose scavenged the free radicals in a dose dependent manner. The result showed a significant scavenging effect of Floragenist Rose by 93% at 0.09% (SEM 1.6%, p<0.01 *vs* UC normalized at 100, SEM 0.4%). (Fig. 13).

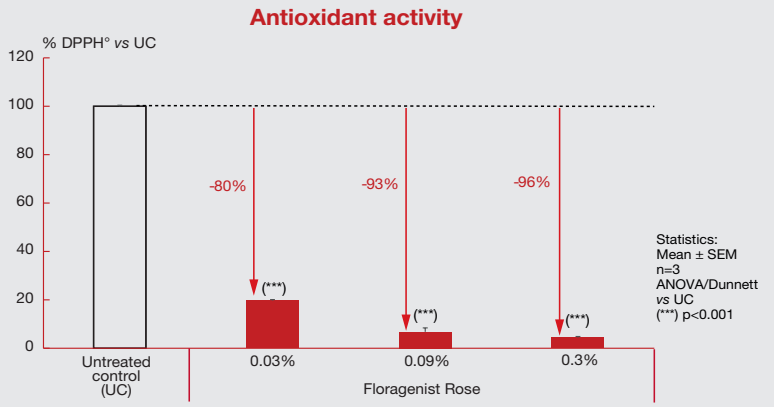


Figure 13 - Effect of Floragenist Rose on inhibition of DPPH oxidation tested with 0.03% to 0.3% of Floragenist Rose.

Antioxidant protection from environmental stress - UVA (*in vitro*)

Free fatty acids and those incorporated into lipids such as in cell membranes are readily oxidized by free radicals, leading to cytotoxic peroxides [Gaschler and Stockwell 2017]. The level of oxidative stress in human keratinocytes exposed to UVA was evaluated by measurement of the end-product malondialdehyde (MDA).
Floragenist Rose helps to protect from UVA induced oxidative stress by decreasing the lipidic peroxidation damage marker MDA (Malondialdehyde) production by 53% at 0.09 of Floragenist Rose (± SEM 2.2%, p<0.001, *vs* UVA stress normalized at 100%, SEM 1.6%), offering a protective effect of 65% against premature skin aging (Fig. 14).

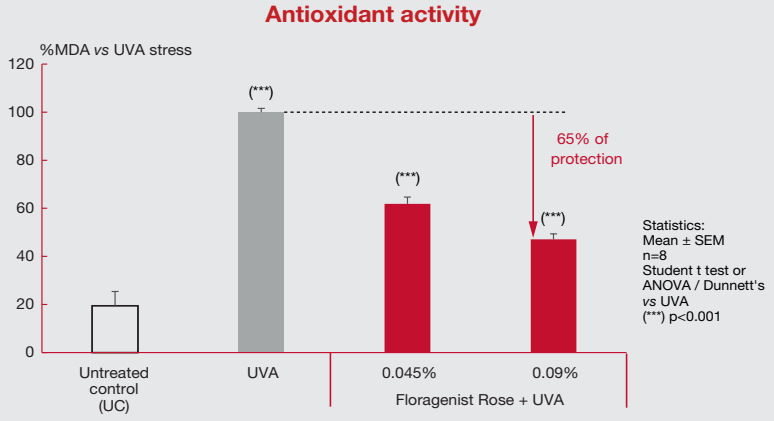


Figure 14 - Inhibition of oxidative stress (released MDA) induced by UVA on human keratinocytes with Floragenist Rose.

CONCLUSION

In vitro and in tubo, Floragenist Rose reduces Free Radical Scavenging (DPPH) while decreases lipid peroxidation induced by UVA on keratinocytes and thus may contribute to neutralizing lifestyle-induced oxidative stress helping to harmonize skin tone.

MATERIALS&METHODS

Studies number

EMB161102-2022-707-DPPH - EMB161102-2023-XXX-CC

Tested products

Freeze-dried coconut-water-rose Extract (batch CS220321.01; INCI name: Cocos Nucifera (Coconut) Fruit Extract (and) Rosa Damascena Flower Extract) was used for the 2 tests.

For DPPH assay, the product was tested diluted at concentrations corresponding to Floragenist Rose from 0.03% to 0.3%. Vitamin E at 0.001% was used to validate the test (data not shown)

For MDA, the product was tested diluted in culture medium at concentrations corresponding to Floragenist Rose of 0.045% and 0.09%. Vitamin E 0.001% was used to validate the test (data not shown).

DPPH radical scavenging assay

The free radical scavenging properties of Floragenist Rose were evaluated in an *in vitro* test with 1,1-diphenyl 2-picrylhydrazyl, also called DPPH. The oxidative form DPPH is stable in solution and can scavenge an electron or a hydrogen radical to become reduced DPPH. DPPH solution was prepared with ethanol and has a purple color. In presence of antioxidant the solution becomes yellow. Absorbance can then be measured at 530nm. The discoloration of DPPH solution reflects free radical scavenging or antioxidant activity.

MDA measurement

The oxidative stress generated by UVA exposure has been measured by recording the levels of released MalonDiAldehyde (MDA), a hallmark of cell membrane lipid oxidation. Human keratinocytes HaCaT cell line, were seeded within standard medium containing FCS and incubated at 37°C, CO_2 5%, relative humidity >95% for 3 days in order to obtain a saturated cell layer. The cell culture medium was exchanged to a standard medium (DMEM) containing products and incubated for 1 day at 37°C. The cell cultures were irradiated by a dose of UVA (20 J/cm²) and then the quantity of released MDA was measured by the thiobarbituric acid (TBA) method.

Results and statistics

For DPPH assay, results are expressed as a percentage referring to UC condition and presented as a Mean ± SEM made in triplicate. The statistics are based on ANOVA One Way and “a posteriori” tests according to Sigmaplot software analysis.

For MDA assay, results are expressed as a percentage referring to UVA irradiated condition and presented as a Mean ± SEM from 2 assays made in quadruplicate (n=8). The statistics are based on Student t test or One Way ANOVA and “a posteriori” tests according to Sigmaplot software analysis. Threshold of significance was set at 5% ($p<0.05$).

EFFICACY

Anti-inflammatory protection from environmental stress (UVB)

In vitro

OBJECTIVE

UVB radiation is a primary cause of skin damage, inducing erythema, an inflammatory skin response characterized by redness and swelling [Putthong *et al*, 2024]. It triggers the release of inflammatory mediators including interleukin-8 (IL-8), a potent inflammatory chemokine contributing to skin redness and inflammation.
The aim of the study was to evaluate the anti-inflammatory performance of Floragenist Rose through the measurement of the release of the cytokine IL-8 by human keratinocytes stressed by UVB exposure.

RESULTS & DISCUSSION

The effect of Floragenist Rose on the release of the pro-inflammatory cytokine interleukin-8 (IL-8) in human epidermal keratinocytes was evaluated by ELISA.

UVB at 50 mJ/cm² significantly induced IL-8 release in human keratinocytes (p<0.001).

Compared to the UVB stress, Floragenist Rose reduced the production of IL8 induced by UVB exposure.

Floragenist at 0.045% significantly decrease the IL-8 release (by 47%, p<0.001), highlighting the robust anti-inflammatory protection achieved. This demonstrates a 100% protection against UVB-induced inflammation, with performance normalized to the UVB stress condition at 100% (Fig. 15).

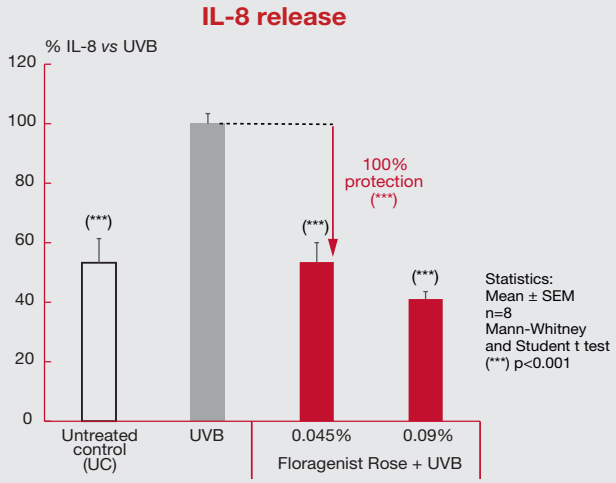


Figure 15 - Quantification of IL-8 synthesis from UVB-treated human keratinocytes with or without Floragenist Rose.

CONCLUSION

By decreasing IL-8 release in keratinocytes stressed by UVB, Floragenist Rose helps to counteract the pro-inflammatory response and thus may contribute to decrease skin redness and harmonize skin tone.

MATERIALS&METHODS

Study number
EMB161102-2023-842-CC

Tested products
Freeze-dried coconut-water-rose Extract (batch CS220321.01; INCI name: *Cocos Nucifera* (Coconut) Fruit Extract (and) Rosa Damascena Flower Extract) was tested diluted in culture medium at concentrations corresponding to Floragenist Rose at 0.045% and 0.09%.

IL-8 quantification
Normal human keratinocytes (NHEK), obtained from healthy donors, were seeded in 48-well plates precoated with collagen I at a rate of 35,000 cells/cm² and cultured in DMEM/Ham's F-12 medium, supplemented with 10% FCS, 1% antibiotic and a cocktail of supplements for 24h at 37°C, under 5% CO_2. Then cells were grown for 72h in MCDB153 medium supplemented with 2% FCS and 1% antibiotic. After, cells were pre-incubated during 24h with the products or with positive control or without any product in MCDB153 supplemented with 1% antibiotic. The medium was then discarded, and HBSS was added before UVB irradiation at 50mJ/cm². After irradiation, cells were re-incubated with the products or with positive control or without any product in MCDB153 supplemented with 1% antibiotic for 24h.

The lack of cytotoxicity of the treatments was checked thanks to the Thiazolyl Blue Tetrazolium Bromide (MTT) reagent. Briefly, after washing, cells were incubated with MTT at 0.25 mg/mL for 2h30 at 37°C. Then, the cells were washed and incubated with 1mL of Dimethylsulfoxide (DMSO). The absorbance was read at 540nm (data not shown).

Finally, the IL-8 released in supernatant medium was measured by a specific commercial ELISA method. Briefly, 100 µL of assay diluent is first added to the wells of the ELISA plate, followed by 50 µL of standards, controls, or samples. Then, the plate is incubated at room temperature for 2 hours. After washing, 100 µL of conjugate is added before the incubation of the plate for 1 hour. After washing, 200 µL of substrate solution is added and the plate is incubated in the dark for 30 minutes. The reaction is stopped by adding 50 µL of stop solution, and absorbance is read at 450 nm.

Results and statistics
Results are expressed as a percentage referring to UC condition and presented as a Mean ± SEM (n=8). A cytoprotective index was also calculated as the % of (UVB Control – treated) / (UVB control – basal).
The statistics are based on Mann-Whitney (MW) Rank Sum test or Student t test with a threshold of significance set at 5% (p<0.05).

EFFICACY

Improvement of skin perfection: redness, luminosity and overall skin homogeneity and health

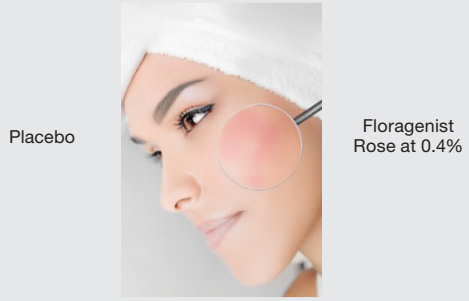
In vivo

OBJECTIVE

"Skin perfection" mainly refers to "skin radiance" and "skin glow" to offer smoothness, clarity, and even skin tone. They are associated with a healthy and attractive appearance, influenced by skin's structural integrity, antioxidant defenses and to some extend skin redness. Today, consumers seek products that enhance skin quality and health, particularly in terms of luminosity and tone evenness.

The aim of this clinical study was to evaluate the skin imperfection reduction efficacy of Floragenist rose at 0.4% as compared to a placebo (Fig. 16).

Double-blind, placebo controlled, split-face and randomized study
on 24 female subjects aged 25-50 years considering having reactive skin



The efficacy of Floragenist Rose at 0.4% was evaluated after 28 days of application using VISIA Imaging and VAESTRO image analysis (luminosity and redness), clinical grading (skin health and homogeneity), and consumer-perception questionnaires (luminosity)

Figure 16 - Clinical protocol.

RESULTS & DISCUSSION

Effect on skin color and luminosity

After 28 days of application, Floragenist Rose at 0.4% showed a significant increase ($p<0.05$) of 0.75 (+1.28%) in luminosity (L*) as compared to baseline, and a significant increase ($p<0.05$) of 0.47 (+0.8%, 2.7x) in luminosity (L*) as compared to placebo (Fig. 17).

After 28 days of application Floragenist Rose at 0.4% showed a significant improvement ($p<0.001$) of 21% in redness (Δa*) as compared to baseline, as well as a significant improvement ($p<0.05$) of 22% in redness (Δa*) as compared to placebo (Fig. 18). An improvement in facial overall Skin homogeneity and a decrease in cheek redness after 28 days of application of Floragenist Rose at 0.4% can also be seen in the illustrative images in Figure 18.

These results demonstrate the ability of Floragenist Rose at 0.4% to improve facial luminosity and redness after 28 days of application.

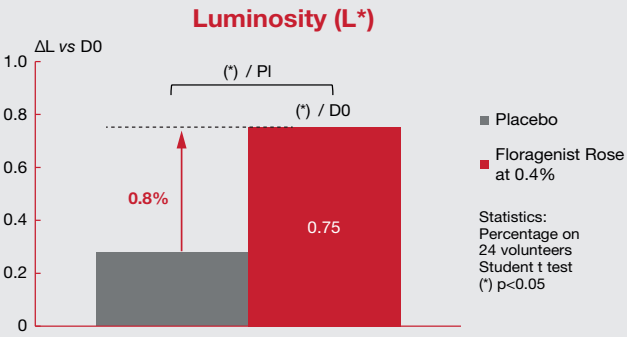


Figure 17 - Measurement of the change in Luminosity (L*) over time after 28 days of Floragenist Rose application.

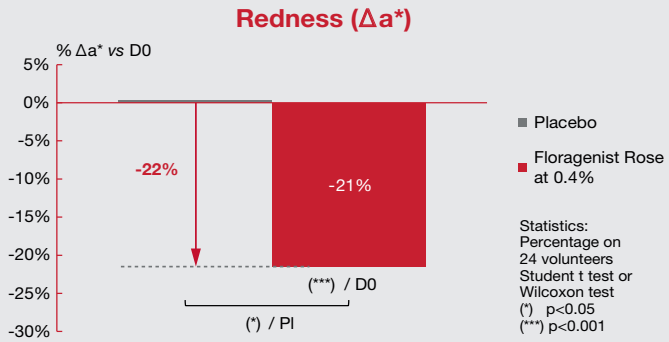


Figure 18 - Measurement of the change in Redness (Δa*) over time after 28 days of Floragenist Rose application.

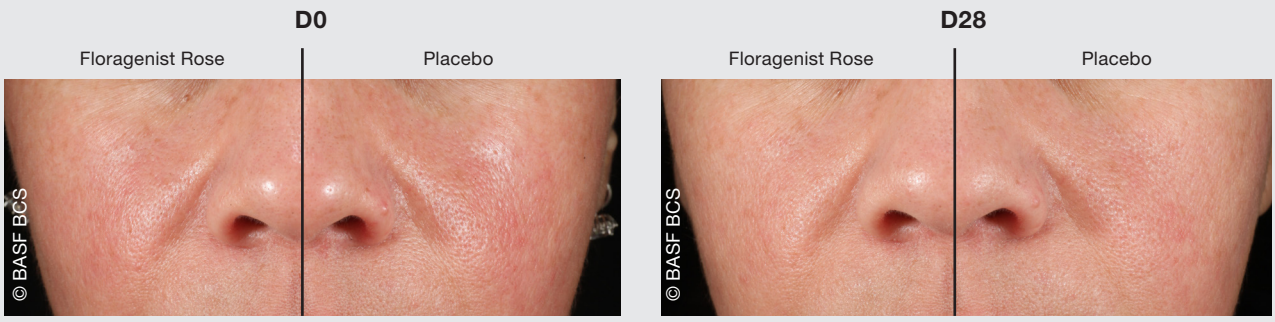


Figure 19 - Illustrative image of change in skin redness and homogeneity over time (Vol 23).

In vivo

Effect on homogeneity & healthy skin by clinical grading

After 28 days of application, Floragenist Rose at 0.4% showed a significant increase ($p<0.01$) of 12% in homogeneity clinical grading, as compared to baseline (Fig. 20). After 28 days of application, Floragenist Rose at 0.4% also showed a significant increase ($p<0.05$) of 7.2% in healthy skin clinical grading, as compared to baseline (Fig. 21). These results demonstrate the ability of Floragenist Rose at 0.4% to contribute to the appearance of skin homogeneity and overall skin health after 28 days of application.

Skin homogeneity (clinical grading)

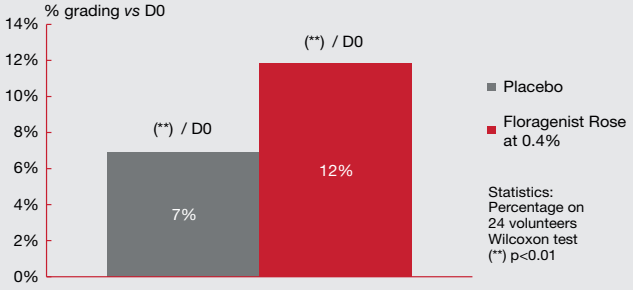


Figure 20 - Change in overall skin homogeneity clinical grading over time.

Healthy skin (clinical grading)

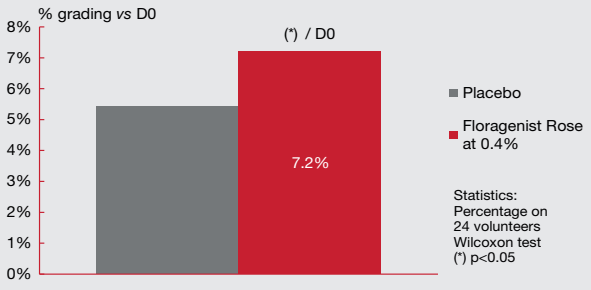


Figure 21 - Change in healthy skin clinical grading over time.

Self-assessment by volunteers

After 28 days of application, a significant majority of the volunteers ($p<0.05$) at 71% agreed that Floragenist Rose made their skin appear luminous. A significant majority of the volunteers did not perceive this benefit with use of the placebo.

CONCLUSION

Floragenist Rose at 0.4% shows the ability to improve skin luminosity and redness after 28 days of application, as well as to contribute to improving overall skin homogeneity and the appearance of overall skin health.

MATERIALS&METHODS

Study number

EMB161102

Tested products

Floragenist Rose at 0.4% (batch PA220322.01)

Study design

The clinical study was a randomized, double-blind, split-face study vs placebo. The efficacy of the emulsion containing Floragenist Rose at 0.4% was compared to the baseline (before treatment, D0) and to the placebo emulsion (similar formulation without Floragenist Rose). The clinical formulations are described in Annex 2. The study was conducted during a period of 28 days with check points at D0 and D28.

Inclusion criteria

24 female subjects aged 25-50 years which consider themselves to have dry and/or reactive/reaction-prone skin and have diffused redness, clear and thin skin.

Application modality

Volunteers applied twice a day, once in the morning and once in the evening, for 28 days the Floragenist Rose emulsion at 0.4% or the Placebo emulsion on hemi-face, under normal conditions of use. The last application was done on the evening of D27 before the visit on D28.

Evaluation methods

VISIA Imaging and image analysis

The VISIA has a multi-point positioning system and ghost imaging capability that captures well registered images to document progress over time. The VISIA captures and automatically analyzes left, right, and frontal facial views. The multi-spectral lighting includes IntelliFlash, cross-polarized and UV lighting to record and measure the surface and subsurface skin conditions. The RBX technology separates the unique color signatures of Red and Brown skin components for unequal visualization of conditions such as spider veins, hyperpigmentation, rosacea, and acne.

Visia images of the face were analyzed with Canfield Vaestro software for skin luminosity (L*) and redness (skin redness difference, expressed as Δa* between the red area and adjacent normal skin).

To calculate the skin redness (Δa*), the following measurements and calculation were done:

(a* red zone – a* normal skin)

Where:

a* red zone = the a* color value of a selected area of redness on the cheek

a* normal skin= the a* color value of a selected area of facial skin outside of the red area of the cheek, near the jaw, representing the base color of the facial skin without excess redness.

Interpretation

An increase of L* means more luminous skin.

A reduction of Δa* means that the red color of the red area is closer to the color of the normal skin, indicating that there is less redness in the affected region and overall, a more uniform skin tone.

Clinical assessment

During the prescreening visit, clinical scientists will evaluate the overall skin conditions on the face, including erythema, psoriasis, sunburn, cuts, scar tissue, abrasion, lesion or other abnormal condition or skin disease, as well as for subjects with diffused redness and clear & thin skin.

During study visits, clinical scientists will grade subjects' facial skin overall Skin homogeneity and healthy skin using VAS scales:

- skin homogeneity: 11-point scale from 0-10, with 0 being the least homogeneous and 10 being the most,
- skin health: 11-point scale from 0-10, with 0 being the least healthy and 10 being the most.

Interpretation: an increase of grading values respectively means an increase of overall skin homogeneity and in the appearance of healthy skin.

Consumer self-perception questionnaire

The volunteers completed self-administered questionnaires to evaluate their perception of the test products' performance for skin luminosity. Questionnaires were completed through Internet-based software at D28. Possible answers were (1) Disagree Strongly, (2) Disagree Somewhat, (3) Disagree Slightly, (4) Agree Slightly, (5) Agree Somewhat, (6) Agree Strongly.

Results and statistics

Instrumental values

Results are expressed as the mean difference between D28 and D0 baseline measurement (D28-D0) for L* and as the mean percentage of change (D28-D0)/D0 for Δa*. The comparisons between products are expressed as difference between their percentage of changes over baseline. All data have been analyzed to confirm the normality of the distribution using the Shapiro-Wilk Test. The following tests were used to compare the change in parameters:

- for those comparisons where the normality of both data sets was validated, either a paired t-or a t-test was used.
- if there was no validation of the normality, or if data was generated by clinical grading, then a Wilcoxon Signed Rank Test or a Mann-Whitney Rank test was used.

Consumer self-perception questionnaire

The analysis involves establishing frequency tables that take into account the number of responses and calculating the frequency of the different possible answers (given as percentage) to each qualitative question.

To evaluate the efficacy and the appreciation of the products for each item, two percentages Z1 and Z2 are calculated as follows:
Z1 = favorable opinion (agree).
Z2 = unfavorable opinion (disagree).

The statistical difference in frequencies (%) between favorable (agreement) and unfavorable (disagreement) opinions is evaluated using Two-tailed binomial test at 5%.



GENERAL CONCLUSION

Floragenist Rose: the skin perfection elixir

Next-generation radiance active that helps imperfect skin regain visible skin perfection by improving skin luminosity, homogeneity in regards to redness as well as the appearance of overall skin homogeneity.

Sourced from Morocco and re-engineered with BASF's proprietary AquaGenesis extraction technology, Floragenist Rose reinvents an iconic flower into a boosted, high performance elixir.

Today's consumers seek both the assurance of authentic botanical ingredients, whose effectiveness is grounded in their natural origin, and the certainty of results verified through rigorous clinical studies. This dual demand reflects a growing desire to combine safety, traceability, and scientifically demonstrated performance.

Skin perfection is an intemporal quest for multifactorial and sensorial skin radiance benefit that refer to skin homogeneity, redness, and luminosity which contribute to a healthier and more attractive appearance.

Floragenist Rose is a potent natural extract of rose flower sourced from Morocco. Powered by AquaGenesis technology, using coconut water as a natural extraction medium, it concentrates key rose phytochemicals beyond standard water extraction, notably delivering +30% flavonol glycosides and 6x more tannins.

In vitro, Floragenist Rose demonstrated significant efficacy by stimulating the gene expression of key epidermal differentiation and cohesion proteins involved in skin optics, such as involucrin, filaggrin, and corneodesmosin, through the epigenetic modulation of miR-203. In parallel, it upregulated twelve genes encoding proteins essential to cell–cell adhesion complexes, adherens junctions , tight junctions, gap junctions, and desmosomes, supporting epidermal structure and cohesion. Moreover, Floragenist Rose exhibited anti-oxidative properties for the defense of keratinocytes against UVA-induced lipid peroxidation and decreased UVB-mediated pro-inflammatory response.

Floragenist Rose was tested at the dose of 0.4% in a placebo-controlled clinical study on 24 volunteers. After 28 days of application, it improved skin luminosity by 2.7x, reduced redness by 22% *vs* placebo, and contributed to improving overall skin homogeneity and the appearance of overall skin health. After 28 days, 71% of volunteers reported a more luminous-looking skin.

Developed to meet the demand for safe, traceable botanicals with demonstrated efficacy, Floragenist Rose combined an iconic botanical source with AquaGenesis technology and epigenetic expertise to support radiance-related skin benefits and a refined "Diamond skin" appearance.

ANNEXES

Annex 1 - Technical data - Available upon request

- Quality and regulatory product information
- Toxicological abstract
- Composition data sheet
- Specifications
- Product Purity Profile
- Formulation Data Sheet
- Natural content (according ISO 16128)
- Botanical sourcing information

Annex 2 - Clinical test formula

Trade name	INCI name	Function	Placebo %	Floragenist Rose %
Water	Water	Solvent	73.99	73.59
Eumulgin SG	Sodium Stearoyl Glutamate	Emulsifier	0.50	0.50
Glycerin 99-5	Glycerin	Humectant	1.00	1.00
Euxyl K 903 (Ashland)	Benzyl Alcohol, Benzoic Acid, Dehydroacetic Acid, Tocopherol	Preservative	1.20	1.20
Emulgade PL 68/50	Cetearyl Glucoside, Cetearyl Alcohol	Emulsifier	5.00	5.00
Lanette O	Hydrogenated Vegetable Glycerides	Consistency agent	1.50	1.50
Cutina HVG	Cetyl PEG/PPG-10/1 Dimethicone	Consistency agent	1.50	1.50
Cetiol RLF	Caprylyl Caprylate/Caprate	Emulsifier	2.00	2.00
Cetiol C 5C	Coco-Caprylate/Caprate	Emulsifier	5.00	5.00
Myritol 318	Caprylic/Capric Triglyceride	Emulsifier	3.00	3.00
Cegesoft PS 6	Olus Oil [EU], Vegetable Oil [CTFA]	Emulsifier	2.00	2.00
Verdessence Xanthan	Xanthan Gum	Rheology modifier	0.25	0.25
Cetiol iSan	Tridecane	Emulsifier	3.00	3.00
Floragenist Rose BC10181	-	Active ingredient	0.00	0.40
NaOH Solution 16%	Sodium Hydroxide, water	pH ajustement	0.06	0.06

Annex 3 - Formulation examples

Rose veil powder cleanser (SC-FR-26-CT-0139-001)

Phase	Ingredients	INCI	% by weight	Function	Manufacturer
A	Verdessence® RiceTouch	Oryza Sativa (Rice) Starch	55.20	Skin Feel Modifier	
A	Lipofructyl® Argan LS 9779	Argania Spinosa (Argan) Kernel Oil	2.00	Active Ingredient	
B	Perfume "Pétales de Rose"	Parfum	1.40	Perfume / Fragrance	AROMA ZONE
C	Texapon® SFA UP Powder	Disodium 2-Sulfolaurate	40.00	Surfactant	
C	Verdessence® Xanthan	Xanthan Gum	1.00	Rheology Modifier	
C	Floragenist™ Rose BC10181	Maltodextrin, Cocos Nucifera (Coconut) Fruit Extract, Rosa Damascena Flower Extract	0.40	Active Ingredient	

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